

1. Project Title

E-Commerce Sales Performance & Customer Insights Dashboard

2. Business Problem

E-commerce businesses generate a large amount of order and customer data. Looking at the raw data makes it difficult to understand sales performance, customer buying behavior, product popularity, delivery time, and customer satisfaction.

The purpose of this project is to organize the data and create an interactive dashboard that makes these areas easier to understand.

3. Project Objective

The main objective of this project is to create an interactive Excel dashboard that helps users quickly understand:

- Overall sales and order performance
- Customer buying behavior
- Popular products
- Customer location
- Delivery performance
- Customer satisfaction
- Recent sales/order trends

The dashboard allows users to apply filters and explore the data dynamically.

4. Dataset Description

The project uses e-commerce order data containing information related to customers, orders, products, quantity, sales amount, ratings, delivery time, gender, order mode, and customer location.

The data is used to analyze both business performance and customer behavior.

5. Data Preparation

Before creating the dashboard, the data was organized so that it could be used for analysis and visualization.

The preparation process included making the data suitable for:

- Pivot Tables
- Pivot Charts
- KPI calculations
- Filtering
- Dashboard visualizations

The prepared data was then used to create the analytical views shown in the dashboard.

6. Business Questions

The dashboard was designed to answer the following business questions:

Sales Performance

- How many total orders were placed?
- How many products were sold?
- What is the total sales amount?
- What has been the recent order trend?

Customer Behavior

- Which channel do customers prefer for purchasing?
- How many products do customers usually buy?
- What is the overall gender distribution?

Product Analysis

- Which products are the most popular?

Geographic Analysis

- Where do the customers live?

Delivery Analysis

- How many days does it normally take to deliver orders?

Customer Satisfaction

- How satisfied are customers based on their ratings?
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7. KPIs

The dashboard contains five main KPIs.

KPI	Current Dashboard Value
Total Orders	2,400
Total Quantity	11,997
Total Amount	\$649.0K
Average Rating	2.3
Average Delivery	4.0 days

These KPIs provide a quick overview of overall business performance.

8. Analysis

8.1 Recent Order Trend

A trend chart is used to show performance over the last 13 weeks.

This helps identify changes in order volume and sales over time.

8.2 Customer Purchase Channel

The dashboard compares different order channels such as:

- App
- Instagram
- Partner App
- Target.com
- Website

This helps understand which channels customers use to place orders.

8.3 Purchase Quantity

A chart shows how many products customers usually purchase in an order.

The categories include:

- 1
- 2
- 3
- 4
- 5
- 6 to 10
- 11+

This helps understand customer purchasing patterns.

8.4 Product Popularity

The dashboard compares products based on their purchase volume.

Products such as T-Shirts, Jeans, Sneakers, Tank Tops, Bikinis, Shorts, and other products are included in the analysis.

This helps identify products with higher customer demand.

8.5 Gender Analysis

The dashboard shows the distribution of customers by:

- Female
- Male
- Other
- Unknown

It also compares gender distribution across different purchasing channels.

8.6 Customer Location

A map is used to show where customers are located.

This helps identify geographic areas with higher customer activity.

8.7 Delivery Performance

A chart shows how many days orders take to be delivered.

This helps evaluate delivery performance and identify the most common delivery time.

8.8 Customer Satisfaction

Customer ratings are visualized to understand how customers rate their experience.

The dashboard can also show rating patterns across different months.

9. Dashboard

The final dashboard combines KPIs, charts, tables, filters, and a map into one interactive report.

The dashboard contains:

- KPI cards
- Weekly trend chart
- Order-mode analysis
- Purchase quantity analysis
- Product popularity analysis
- Gender analysis
- Customer location map
- Delivery-time analysis
- Customer-rating analysis

The dashboard also includes interactive filters such as:

- Gender
- Order Mode

These filters allow users to change the analysis dynamically.

10. Key Insights

Based on the current dashboard view, several observations can be made.

Overall Performance

The dashboard shows:

- **2,400 total orders**
- **11,997 total quantity sold**
- **\$649.0K total amount**
- **2.3 average customer rating**
- **4.0 average delivery time**

Customer Behavior

Customers use multiple channels to place orders, including the App, Website, Instagram, Partner App, and Target.com.

The dashboard allows the performance of these channels to be compared.

Product Demand

The product analysis shows that some products have much higher purchase volumes than others, which helps identify popular products.

Delivery

The delivery chart shows that most orders are delivered within the lower range of the delivery-day categories, with fewer orders taking much longer.

Customer Satisfaction

The average rating shown on the dashboard is **2.3**, which indicates that customer satisfaction is an important area to investigate further.

These observations should be validated with the underlying data before making business decisions.

11. Recommendations

Based on the dashboard, the following areas can be considered for further business analysis:

1. Investigate the reason for the relatively low average customer rating.
2. Study the factors behind longer delivery times.

3. Focus inventory planning on products with consistently high demand.
4. Compare customer behavior across different order channels.
5. Analyze geographic demand to understand where customers are concentrated.
6. Monitor weekly order trends to identify periods of increasing or decreasing demand.

These recommendations should be refined after deeper analysis of the underlying data.

12. Tools & Techniques

Tools

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- Excel Dashboard

Techniques

- Data summarization
 - KPI analysis
 - Trend analysis
 - Customer segmentation
 - Product analysis
 - Geographic analysis
 - Delivery analysis
 - Customer satisfaction analysis
 - Interactive filtering
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13. Conclusion

This project demonstrates how Excel can be used to convert e-commerce order data into an interactive analytical dashboard.

The dashboard brings together sales performance, customer behavior, product popularity, customer location, delivery performance, and customer satisfaction in one place.

The main learning from this project was not only creating charts and Pivot Tables, but also understanding how different business questions can be answered using data.

This project is my first step toward building more advanced analytics projects using **SQL, Power BI, and DAX.**